

Manual

Bailey Crawford

Overview

This tool evaluates individual fairness in a trained machine learning model. It identifies cases where two similar inputs produce different classifications when only a sensitive feature changes. These cases are called individual discriminatory instances (IDIs).

The project includes:

- `fairness_tool.py` – improved method
- `lab4.py` – baseline comparison

Setup

Ensure the following files are in place:

- `model/processed_kdd.csv`
- `model/model_processed_kdd.h5`

Install dependencies:

```
pip install -r requirements.txt
```

Running the Proposed Method

Run:

```
python fairness_tool.py
```

This will:

- Generate valid sample pairs
- Force sensitive feature changes
- Remove duplicate pairs
- Compute IDI ratios
- Save results to CSV files
- Perform Wilcoxon statistical tests

Running the Baseline

Run:

```
python lab4.py
```

This produces:

- `baseline_results.csv`
- `baseline_summary.csv`

Generating the Graph

Run:

```
python plot_graph.py
```

This creates:

```
idi\_vs\_budget.png
```

Outputs

- fairness_results.csv
- fairness_summary.csv
- wilcoxon_results.csv
- baseline_results.csv
- baseline_summary.csv
- idi_vs_budget.png

Interpretation

- Higher IDI ratio indicates more detected fairness issues
- Lower standard deviation indicates more stable results
- $p < 0.05$ indicates statistically significant differences